MECH 421	Mechanical Design I	3(3,0)
Engineering materials properties; tools of machine design; stress, strain, and deformation of machine parts; buckling of columns; statistical methods and safety consideration; static and fatigue failure theories; design of machine parts for strength under steady loads; design of machine parts for rigidity, deflection of beams and rotation of shafts; design of machine parts for strength under cyclic/dynamic loads; codes and standards. **Prerequisite**: MECH 320. Annually.		

MECH 422	Manufacturing Processes	3(3,0)
General overview of manufacturing; properties of engineering materials; steelmaking; dimensions and tolerances; manufacturing processes: rolling, forging, extrusion, powder metallurgy; mechanical assembly: welding, brazing, soldering, bonding; welding QA/QC; non-destructive testing (NDT); machining; casting; sheet metal forming; plastic injection molding; polymers and composites. **Prerequisite**: MECH 320. Annually.		

MECH 423	Mechanical Design II	3(3,0)
Static & fatigue design of shafts and axles; design of permanent and non-permanent joints (screws, fasteners, bolting, welding); rolling-contact bearings; lubrication and journal bearings; Gears-general, spur, and helical gears; flexible mechanical elements; power transmission case study; material selection; codes and standards; computer modeling; static and fatigue design using the principles of applied fracture mechanics combined with finite elements analysis results; failure analysis; team project. **Prerequisites**: MECH 223, MECH 321, and MECH 421. Annually.		

MECH 431	Heat Transfer	3(3,0)
The course introduces the relationship between heat transfer and thermodynamics; steady and transient heat conduction in one and two dimensions; analytical and numerical solutions; extended surfaces; internal and external flow; forced and natural convection; and an introduction to heat radiation. **Prerequisite**: MECH 333 or MECH 335 and MATH 317. Annually.		

MECH 431L	Heat Transfer Lab	1(0,2)
This laboratory course provides hands-on experience with fundamental heat transfer principles. Students will work with a variety of experimental setups, including a temperature-measurement bench for calibrating and recording thermal data and heat-exchanger units configured for cross-flow, parallel-flow, and mixed-flow operations. Additionally, a thermal conductivity apparatus is used to investigate temperature gradients and calculate conductivity across various solid specimens. **Co-requisite**: MECH 431. Annually.		

MECH 435	HVAC	3(3,0)
Air conditioning basics; occupant comfort and indoor air quality; heating and cooling load calculations; HVAC controls; air conditioning systems and air delivery systems; refrigeration basics, methods, and applications. **Prerequisite**: MECH 431. Annually.		

MECH 499	Co-op Work Experience	3(3,0)
This Co-op work experience is designed to provide students with a five-week short-term work experience in mechanical engineering. Students are encouraged to network with industry professionals in their discipline. A report, a poster, and a PowerPoint presentation are required. **Prerequisites**: ENGL 217; Senior Standing.		

MECH 520	Finite Element Method for Engineers	3(3,0)
This course focuses on the basic concepts of finite element methods; formulating shape functions and basic structural element equations; modeling and analysis of 1-D, 2-D, and 3-D structural analysis and heat transfer problems; finite element modeling and analysis using MATLAB and ANSYS. **Prerequisites**: MECH 320, MATH 421, and MECH 431. Annually.		

MECH 521	Vibrations and Acoustics	3(3,0)
This course covers Free Vibrations, Vibrations with Damping, Harmonically Excited Vibrations, Complex N-Degree-of-Freedom Systems, Continuous Vibrations, Fundamentals of Sound Waves and Acoustics, and Basic Machinery Vibration. Prerequisite: MECA 340. Annually.		

MECH 595A	BE Summative Learning Project I	1(1,0)
A team-oriented project includes conducting and formulating experimental laboratory work and design problems on a specific, approved specialty topic. The project includes a literature review, a scope of work, and a project proposal. **Prerequisite**: Senior Standing, ENGL 217. Annually.		

MECH 595B	BE Summative Learning Project II	3(3,0)
A team-oriented project includes conducting and formulating experimental laboratory work and design problems on a specific, approved specialty topic. The project includes a literature review, design work, data collection, experimentation, data analysis, and technical writing. **Prerequisite**: MECH 595A. Annually.		

II. Elective Courses

Students may meet their technical electives requirements from the following selection of elective courses.

MECH 436	Petroleum Refining Technology for Mechanical Engineers	3(3,0)
Crude oil; petroleum products; refinery configuration; major process equipment; crude distillation; catalytic reforming; hydrotreating; gasoline; gas treating; product blending; risk-based inspection; environmental control; economics. **Prerequisite**s: CHEM 211, MECH 330, and MECH 320. On demand.		

MECH 437	Heat Exchangers Design, Performance, and Inspection	3(3,0)
Quick review of “mechanics of materials” and “heat transfer”; Codes & Standards; Heat exchanger types and applications; Thermal design; Mechanical design;		

Inspection and testing; Efficiency; TEMA standard use; Design project; Laboratory testing. **Prerequisites**: MECH 320 and MECH 431. On demand

MECH 445	**Automotive Engineering**	**3(3,0)**
Studies of automotive components: engine parts (crankshaft, camshaft, inlet/exhaust valves, piston/cylinder mechanism); power boosting (turbocharging/supercharging); transmission; steering mechanism design; engine combustion and emissions; major project on simulation of overall engine performance. **Prerequisites**: MECH 330 or 333 and MECH 321. On demand.		

MECH 448	**Plumbing Engineering**	**3(3,0)**
Soil, waste, and vent piping; drainage system; piping systems for domestic and industrial hot and cold water, fuel oil, de-ionized and de-mineralized water; laboratory compressed air and vacuum; laboratory outlets; laboratory drains; vacuum pumps; general piping installation. **Prerequisites**: MECH 335 or MECH 333. On demand.		

MECH 451	**Fluid Mechanics II**	**3(3,0)**
Review of incompressible flow; potential flow; similitude, dimensional analysis, and modeling; external and open channel flow; pumps and turbines. **Prerequisite**: MECH 335. On demand		

MECH 453	**Buildings and Energy**	**3(3,0)**
Selected topics in low-energy building design: building history and energy use; building physics; architectural and material considerations; building energy systems and the use of renewable energy systems; hands-on project. **Prerequisite**: MECH 330 or MECH 333. On demand.		

MECH 455	**Gas Turbines**	**3(3,0)**
Thermodynamics of gas turbines; Gas turbine design; comparison of designs; shaft power and aircraft propulsion cycles; component efficiencies; basic components; general considerations of turbomachinery design; axial compressors; combustors; clean combustion systems; fuel control; design-point performance calculations; off-design behavior and matching procedure; project. **Prerequisite**: MECH 330 or MECH 333. On demand.		

MECH 461	**Thermal Systems Design**	**3(3,0)**
Modeling of thermal systems; system solution; design applications of optimization methods: Lagrange multipliers, search methods, and dynamic, geometric, and linear programming**. Prerequisite**: MECH 431. On demand.		

MECH 468	**Internal Combustion Engines**	**3(3,0)**
Fuels and their properties; review of basic thermodynamics and gaseous mixtures; combustion thermodynamics; engine types; combustion chambers; valve mechanisms; dynamic balance of ICE; flame temperature; dissociation and detonation; turbocharging and supercharging; testing and measurements; engine		

characteristics and performance; laboratory demonstration. **Prerequisite**: MECH 334. On demand.

MECH 470	Product Design and Development	3(3,0)
Idea generation; concept generation; concept selection; functional analysis; engineering design process for systems and components; economic consideration; reliability analysis; product safety; design project. **Co-requisite**: MECH 421. On demand.		

MECH 471	Refrigeration	3(3,0)
Methods of producing cold: thermodynamic basis, first and second law analyses – vapor compression systems; ideal vs. actual cycles; single-stage, multistage and cascaded systems; environmental considerations and alternate refrigerants; vapor absorption systems. **Prerequisite**: MECH 334. On demand.		

MECH 498	Special Topics in Mechanical Engineering	3(3,0)
Guided study and research on major-related topics. **Prerequisite**: Instructor's Consent. On demand.		

MECH 523	Mechanics of Composite Materials	3(3,0)
Analysis of a thin sheet of aligned-fiber reinforced, elastic matrix (lamina) to layered structures comprised of these sheets (laminates); mechanics of the thick-sectioned components (of many microstructural compositions) desirable in many engineering applications; processing of composite materials; mechanical and thermal properties and their degradation with time; practical use; fatigue versus static strength; finite element application. **Prerequisite**: MECH 320. On demand.		

MECH 524	Elasticity and Plasticity	3(3,0)
Analysis of Cartesian tensors using indicial notation; 3-D stress and strain concepts; point stress and strain about tensor concepts; deviatoric and spherical stress state; Octahedral shear stress; equations of equilibrium and compatibility; constitutive laws for 3-D elastic and elastic-plastic stress state; kinematic & isotropic hardening models in plasticity; finite element applications. **Prerequisite**: MECH 320. On demand.		

MECH 525	Advanced Dynamics	3(3,0)
General introduction; rigid body kinematics: translation (3-D), rotation (3-D), compositions of rotations, Euler-angles presentation of rotation, Roll-pitch-yaw presentation of rotation, axis-angle presentation of rotation, homogenous transformations, rotation matrix derivative, angular velocity, linear velocity, acceleration; rigid body kinetics: review of Newtonian dynamics, generalized coordinates, holonomic and non-holonomic constraints, virtual displacements, virtual work, D'Alembert's principle, Lagrange equations, Hamilton's principle, canonical transformations; applications of Hamiltonian dynamics; multi-body dynamics. **Prerequisite**: MECH 220. On demand.		

MECH 527	Modern Material Science	3(3,0)
Advanced coverage of metals and alloys; applications and processing of ceramics; characteristics, applications, and processing of polymers; surface property enhancement; materials simulation; economic, environmental, and societal issues in materials science; biomaterials and nanoscale materials. Prerequisite: MECH 223. On demand		

MECH 530	Computational Fluid Dynamics	3(3,0)
This course introduces computational techniques to solve thermal-fluid problems. It integrates applications using ANSYS and CFD theory where it covers the following: Structured and unstructured grids; an overview of various discretization methods with emphasis on the finite volume method; Diffusion term discretization; Convection term discretization; Source term discretization; Transient term discretization, implicit and explicit schemes; Direct and Iterative solvers, tridiagonal matrix algorithm (TDMA), Gauss-Seidel; CFD analysis process (domain generation, domain meshing, domain setup or mathematical modeling, and post-processing); customizing Ansys fluent solver through user-defined functions; application of ANSYS code to practical engineering problems. Prerequisite: MECH 431.		

MECH 541	Renewable Energy	3(3,0)
Energy situation; energy conversion; solar power; Photovoltaic and thermoelectric conversion, ocean, tidal, and wave power; wind power; biomass energy; the case of nuclear power; project. **Prerequisite**: MECH 330 or MECH 333. On demand.		

MECH 570	Digital Manufacturing	3(3,0)
In this course, the student will get an insight into diagnosing and correcting operational flaws; modeling and fabrication based on additive manufacturing technologies; automation technologies; production planning, process quality control; use of modern four-axis Computer Numerical Control (CNC) machines, 3D printers, and up-to-date software programs, including AutoCAD, Inventor, SolidWorks, and VisualCam. **Prerequisite**: MECH 422 and MECA 541. On demand.		

MECH 571	Optimization in Engineering Design	3(3,0)
Problem definition, cost function and constraints; local vs. global methods; deterministic vs. stochastic methods; linear vs. non-linear programming; constrained vs. unconstrained; continuous vs. discrete; gradient-based methods; combinatorial optimization techniques; applications to various engineering problems (single/multi-objective); project. **Prerequisite**: MATH 421. On demand.		

III. AI-Restricted Elective Courses

Students must choose at least one elective course (3 credits) from the following selection of AI-Restricted courses.

MECA 445	Advanced Programming for AI & Robotics	3(3,0)
This advanced course, built upon the prerequisite "Mechatronics Systems Programming," delves into the practical application of Python for developing intelligent robotic systems. A significant portion of the course will be dedicated to the Robot Operating System (ROS), providing students with the skills to design, implement, and test complex robotic software. Students will learn core ROS concepts, including nodes, topics, services, and parameters, as well as essential ROS tools for simulation and real-world robot interaction. Furthermore, the course will explore techniques for integrating Artificial Intelligence (AI) into mechatronics applications using Python. This will include an introduction to relevant Python libraries for machine learning, computer vision, and data processing, along with practical examples of how AI can be applied to tasks such as perception, decision-making, and control in robotic systems. The course will emphasize hands-on projects and assignments using platforms such as Raspberry Pi and NVIDIA Jetson Nano to provide students with practical experience in developing and deploying advanced robotic and AI-driven mechatronic solutions. **Prerequisite**: MECA 444L. Annually.		

MECA 535	Expert Systems and Applied Intelligence	3(3,0)
This course explores the core principles of intelligent systems and their application in solving complex engineering and decision-making problems. Students will learn perception methods, reasoning techniques, and search strategies, as well as key AI tools such as neural networks, evolutionary computing, fuzzy logic, and intuitionistic fuzzy logic. The course also introduces intelligent programming using Prolog/python to model knowledge and build rule-based expert systems. Through practical examples, students will gain hands-on experience in developing AI-driven solutions for real-world applications. **Prerequisite**: MECA 444L.		

MECA 536	Applied Deep Learning for Intelligent Systems	3(3,0)
This course introduces the basics of computer vision and machine learning, focusing on their use in intelligent systems. Students will learn techniques for object detection, image understanding, and sensor data interpretation, using deep learning tools to build decision-making models. Hands-on practice with simulation platforms will allow students to design and test vision-based solutions in realistic scenarios. At the end of the course, participants will gain practical skills in combining perception, planning, and control, preparing them for applications in autonomous systems and other technologies. **Prerequisite**: MECA 444L.		

MECA 546	Smart Energy Management Systems	3(3,0)
This course delves into smart energy management systems (SEMS), focusing on versatile operating modes for solar photovoltaic (PV) and battery energy storage (BESS) systems. It equips mechatronics engineering students with the fundamental knowledge and practical skills required to design, implement, and optimize these crucial technologies for a sustainable future. Furthermore, the course discusses the potential of using generative AI for the design and development of smart energy management systems. A capstone case study is presented and analyzed. **Prerequisite**: MECA 442.		

MECH 522	Mechanism Synthesis: Classical Kinematics to AI-Driven Optimization	3(3,0)
This course introduces the theory and computational practice of mechanism synthesis, with emphasis on the design of four-bar linkages for path, motion, and function generation. Classical kinematic synthesis methods are integrated with numerical optimization techniques, including Newton–Raphson, least-squares, and gradient-based methods, as well as intelligent metaheuristic algorithms such as genetic algorithms, tabu search, and ant colony optimization. The course also exposes students to data-driven and AI-assisted approaches, including deep neural networks for inverse mechanism design, as advanced computational tools for automation, design-space exploration, and decision support in mechanism synthesis. Computational projects provide students with hands-on experience in mechanism modeling, simulation, error analysis, optimization, and the synthesis of mechanical systems under practical design constraints. **Prerequisite:** MECH 321.		